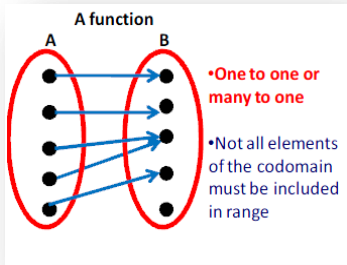


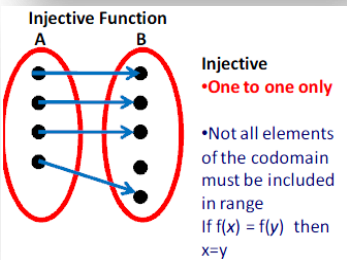
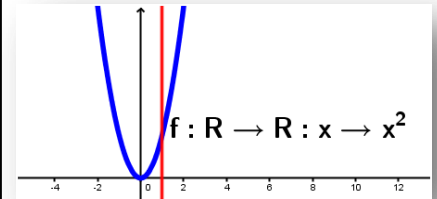
Functions and different types of functions



A relation is a **function** if for every x in the **domain** there is **exactly one** y in the codomain.

A **vertical line** through any element of the domain should intersect the graph of the function exactly once.

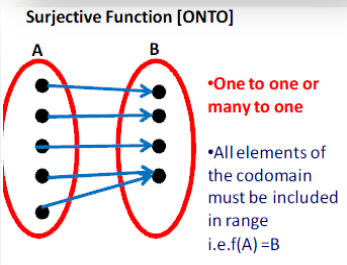
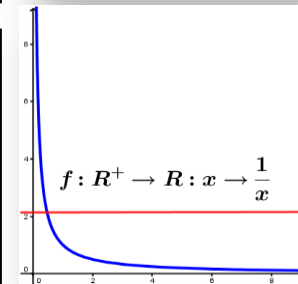
(one to one or many to one but not all the Bs have to be busy)



A function is **injective** if for every y in the **codomain B** there is **at most one** x in the domain.

A **horizontal line** should intersect the graph of the function at most once (i.e. not at all or once).

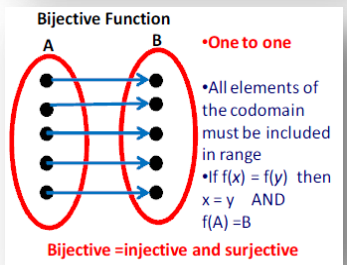
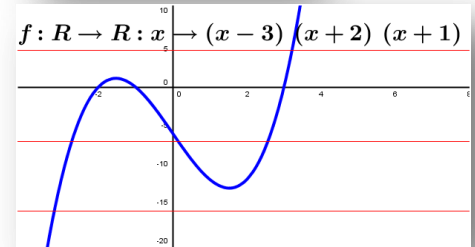
(one to one only but not all the Bs have to be busy)



A function is **surjective** if for every y in the **codomain B** there is **at least one** x in the domain.

A **horizontal line** intersects the graph of the function at least once (i.e. once or more). The range and the codomain are identical.

(one to one or many to one and all the Bs must be busy)



A function is **bijective** if for every y in the codomain there is **exactly one** x in the domain.

A **horizontal line** through any element of the range should intersect the graph of the function exactly once. *(one to one only and all the Bs must be busy).*

